This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2022.Doi Number

A Novel Transformer Model with Multiple Instance Learning for Diabetic Retinopathy Classification

Yaoming Yang[1], Zhili Cai[1], Shuxia Qiu[1,2], and Peng Xu[1,2]

[1]College of Science, China Jiliang University, Hangzhou 310018, China

[2]Key Laboratory of Intelligent Manufacturing Quality Big Data Tracing and Analysis of Zhejiang Province, Hangzhou 310018, P.R. China

Corresponding author: Peng Xu (e-mail: xupeng@cjlu.edu.cn).

"This work was supported by the National Natural Science Foundation of China [grant numbers 51876196]."

ABSTRACT Diabetic retinopathy (DR) is an irreversible fundus retinopathy. A deep learning-based automated DR diagnosis system can save diagnostic time. While Transformer has shown superior performance compared to Convolutional Neural Network (CNN), it typically requires pre-training with large amounts of data. Although Transformer-based DR diagnosis method may alleviate the problem of limited performance on small-scale retinal datasets by loading pre-trained weights, the size of input images is restricted to 224×224. The resolution of retinal images captured by fundus cameras is much higher than 224×224, reducing resolution in training will result in the loss of valuable information. In order to efficiently utilize high-resolution retinal images, a new Transformer model with multiple instance learning (TMIL) is proposed for DR classification. A multiple instance learning approach is firstly applied on the retinal images to segment these high-resolution images into 224×224 image patches. Subsequently, Vision Transformer (ViT) is used to extract features from each patch. Then, Global Instance Computing Block (GICB) is designed to calculate the inter-instance features. After introducing global information from GICB, the features are used to output the classification results. When using high-resolution retinal images, TMIL can load pre-trained weights of Transformer without being affected by weight interpolation on model performance. Experimental results using the APTOS dataset and the Messidor-1 dataset demonstrate that TMIL achieves better classification performance and reduces inference time by 62% compared with that directly inputting high-resolution images into ViT. And TMIL shows highest classification accuracy compared with the current state-of-the-art results. The code will publicly available at https://github.com/CNMaxYang/TMIL.

INDEX TERMS Vision Transformer, Multiple instance learning, Diabetic retinopathy, High-resolution fundus retinal images, Medical image classification

I. INTRODUCTION

Diabetic retinopathy (DR) is a retinal vasculopathy caused by diabetes, has led to a great number of blindness cases among the working-age population worldwide [1]. Since the damage to vision caused by DR is irreversible, early detection and treatment are crucial for preserving DR patients' eyesight. With the rapid development of CNN, it has been successfully applied for the detection and classification of DR [2-4].

Recently, Transformer has shown superior performance to CNN on natural images [5], and therefore has been introduced in the field of medical image [6]. Kumar *et al.* [7] tested CNN, Transformer and MLP on the APTOS dataset, and they found that the DR classification performance of Transformer is better than that of CNN. While the cost of acquiring and annotating medical images is high, and the dataset is limited. Thus, the classification performance of ViT may be weaker than that of CNN due to the lack of inductive biases [5]. Although transfer learning can alleviate this issue, in order to avoid missing pre-trained weights, the resolution of medical images needs to be adjusted to match the resolution of the pre-trained images (e.g., 224×224). However, medical images typically have a higher resolution than the resolution of natural images used for pre-training [8-10]. For DR, ophthalmologists can identify lesion features such as hemorrhages, hard exudates, microaneurysms, and more in high-definition retinal images, as depicted in Figure 1. However, some lesion

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

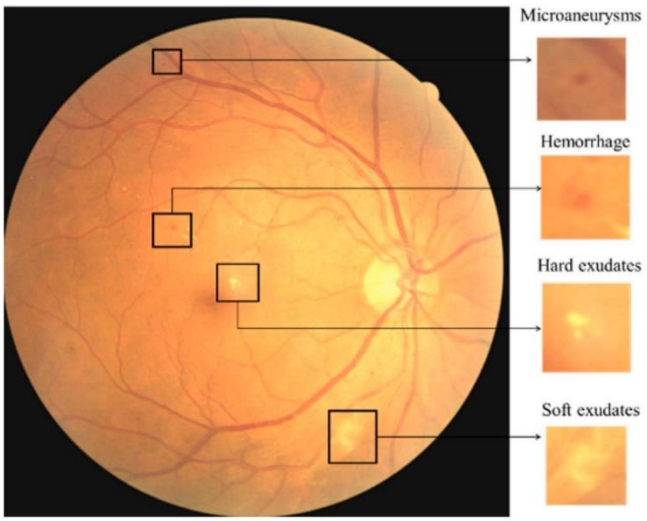


FIGURE 1. A schematic diagram of DR lesion features, sourced from [11].

features are extremely small within the image, as seen in Figure 1, where the microaneurysm lesion feature occupies just a few pixels, making it easy to overlook. Further image compression in such instances is likely to result in the loss of these features, which can have an adverse impact on the model's classification capabilities. Moreover, the information loss caused by compressing medical images is not conducive to alleviating the problem of limited training data, and it also hampers fine-tuning the model on medical image datasets. Furthermore, inputting high-resolution medical images into ViT will lead to a larger sequence length, which significantly increases computational time for the self-attention mechanism.

In order to use higher resolution images while avoiding a significant increase in network inference time, a new Transformer-based multiple instance model (TMIL) is proposed for DR classification in this paper. As illustrated in Figure 2, a single retinal image is treated as a "bag" and the image is uniformly divided into non-overlapping image patches (instances). Then, ViT-Small is used to extract features from each instance, followed by adding positional embeddings. The GICB is proposed to aggregate features from different instances and obtain global features, which are then used to generate the final prediction for the bag. The main contributions of this study are listed as follows:

- A simple and efficient multiple instance model along with a concise GICB is developed for the high-resolution retinal fundus images.
- The model achieves an average accuracy of 85.6%, an average Area Under Curve (AUC) of 95.6%, an average Kappa score of 91.5%, and an average sensitivity of 73.7% for the five-level classification of DR on the APTOS dataset. On the Messidor-1 dataset, the model achieves an average accuracy of 93.1%, an average AUC of 97.4%, and an average sensitivity of 86.9% for the referable DR classification. The result outperforms that directly inputting high-resolution images into ViT, and the accuracy surpasses the state-of-the-art results.
- Compared to the original ViT, the proposed model not only achieves better performance but also significantly reduces inference time by approximately 62%.

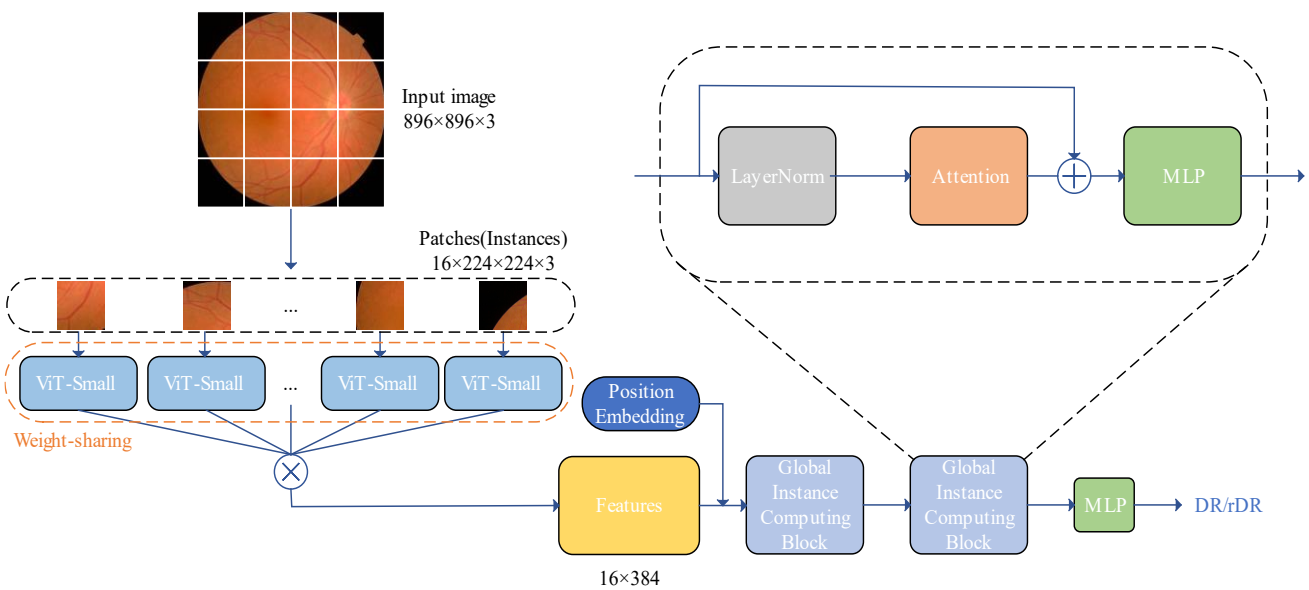


FIGURE 2. The architecture of proposed TMIL for DR classification. The model initially uniformly divides the image into patches (instances) and then employs the weight-shared ViT-Small to extract features for each instance. After feature extraction, these features are input into the GICB to complement global features, and finally, a MLP is utilized to output the classification results.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

II. RELATED WORK

A. TRANSFORMER IN DIABETIC RETINOPATHY DOMAIN

Mohan *et al.* [12] tested networks such as ViT, ResNet101, and Inceptionv3 on IDRiD and EyePACS datasets. They found that ViT outperformed CNN like ResNet101 and Inceptionv3 in terms of classification performance. Meanwhile, Matsoukas *et al.* [13] tested the classification performance of ResNet50 and DeiT-S on APTOS, ISIC, and CBIS-DDSM. They found that ViT pre-trained on ImageNet performs comparably to CNN with limited data. However, ViT outperforms CNN by further pre-training ViT using self-supervised methods based upon ImageNet pre-training. The DR grading and lesion detection are generally treated as separate tasks, which restricts model performance and practical deployment. Sun *et al.* [14] proposed a lesion-aware transformer (LAT) model that can simultaneously perform lesion detection and DR prediction on retinal fundus images. It has been shown to achieve better performance than the state-of-the-art methods for DR grading and lesion detection.

B. TRANSFORMER WITH MULTIPLE INSTANCE LEARNING IN MEDICAL IMAGE

Yu *et al.* [15] proposed a multiple instance model called MIL-VT for the classification of DR. The model builds upon ViT-Small and incorporates a designed "MIL head", and one of the classification result was obtained from the Class Token of ViT-Small. Each of the rest token, except for the Class Token, was treated as an instance and input to the MIL head. A bag prediction was finally output by the MIL head. Myronenko *et al.* [16] applied the multiple instance learning for the classification of histology whole slide images (WSI). They embedded a Transformer block to capture the dependencies among instances because the tumor grading may depend on specific patterns in different locations within the WSI. They also introduced an instance-wise loss function based on instance pseudo-labels and achieved state-of-the-art results on the PANDA challenge dataset. Qian *et al.* [17] presented a model named Swin-MIL based on the Swin Transformer. They also recognized that the independence among instances in MIL restricts the model's performance due to the lack of inter-instance correlation. Therefore, they introduced the Transformer into MIL to capture global dependencies and got state-of-the-art results in weakly supervised segmentation methods on a colon cancer dataset.

III. METHODOLOGY

A. MULTIPLE INSTANCE LEARNING

Multiple instance learning was first introduced by Dietterich *et al.* in 1997 [18] to predict the activity of drug molecules. They took the molecule conformation as an instance and all the conformations of the same molecule as a bag. If any instance in the bag had a class label of 1, the bag was labeled as 1. If all instances in a bag had a class label of 0, the bag was labeled as 0. The ultimate goal of multiple instance learning is to train a multiple instance model that can classify bags. This approach is currently known as the Bag-Space (BS) paradigm [19]. In addition to the BS paradigm, there are also the Instance-Space (IS) paradigm and the Embedded-Space (ES) paradigm [20]. In the IS paradigm, instance predictions are obtained first, followed by bag predictions. While, in the ES paradigm, all instances are embedded into a low-dimensional representation, and bag predictions are generated based on these low-dimensional representations [21]. In this work, the ES paradigm is used to construct the multiple instance model in order to leverage the global information of bags as much as possible.

B. VISION TRANSFORMER

The ViT is a type of Transformer model applied in the field of computer vision. Dosovitskiy *et al.* [5] modified the original Transformer architecture by removing the decoder and fine-tuning the model details to make it suitable for natural images. They named this model as ViT and achieved state-of-the-art classification results on the ImageNet dataset, which outperforms CNN-based approaches.

The present Transformer model is based on ViT-Small [5], and the architecture of ViT-Small is illustrated in Figure 3. The original structure of this model consists of 12 Transformer blocks with an embedding vector dimension of 384. The patch size is 16×16×3, and the input image size is 224×224×3. The model divides the input image into non-overlapping image patches and encodes them into 384-dimensional vectors. After encoding, a learnable position embedding vector of the same dimension is added to these vectors. Then these vectors are input to the Transformer blocks for further encoding. Each Transformer block consists of 2 LayerNorm layers, 1 multi-head attention module, and an MLP with two fully connected layers. Finally, a simple fully connected layer is used to output the classification results.

In the proposed TMIL approach, the fully connected layer used for the final classification in ViT-Small is removed, and the remaining structure is retained for extracting instance features.

C. GLOBAL INSTANCE COMPUTING BLOCK

The instances are obtained through segmentation on the original image, which may result in instances with mild lesion features or without any lesions being present in the bag with severe lesion. In order to avoid the influence of specific instances and utilize the global information of instances within a bag, a relatively simple GICB is designed here to perform computations on the features extracted from different instances. As shown in Figure 2, this module consists of a LayerNorm layer, a multi-head self-attention mechanism, and a single-layer fully connected layer. The multi-head self-attention mechanism is used to compute the information among features extracted from different instances.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

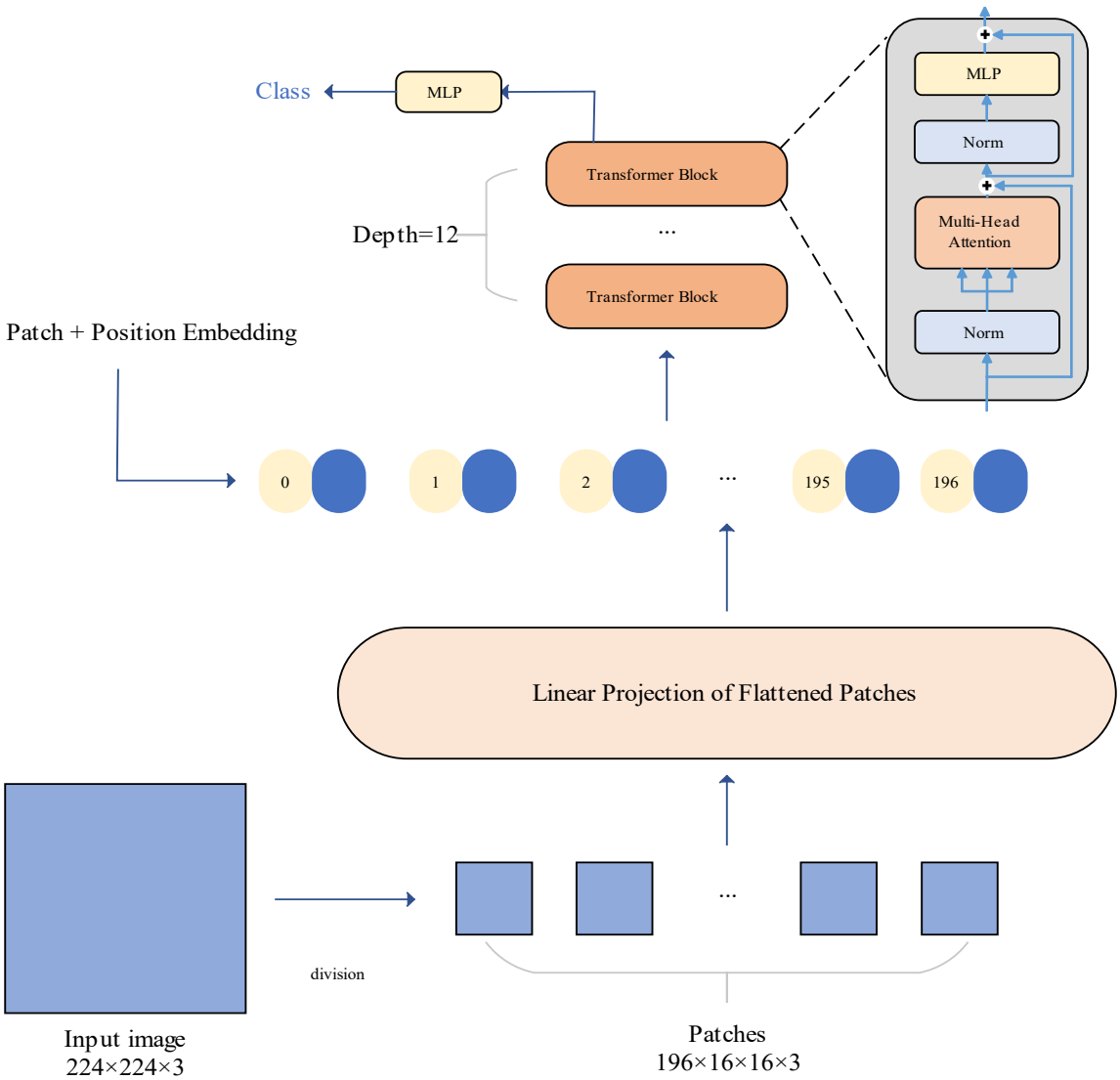


FIGURE 3. **The architecture of ViT-Small.**

And the global information is introduced accordingly. Subsequently, a non-linear transformation is introduced through the fully connected layer. The multi-head self-attention mechanism can be calculated by:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (1)$$

$$\text{MultiHead}(Q,K,V) = Concat(\text{head}_1, \text{head}_2, \dots, \text{head}_\text{h})W^O \qquad (2)$$

Equation (1) represents a standard "Scaled Dot-Product Attention", where Q, K and V are Query Key and Value, respectively. These three components are derived from the output of the previous layer, hence referred to as self-attention. In (2), h represents the number of heads in the multi-head attention mechanism. In the ViT-Small model used in this study, the value of h is set to be 6. The value of a single head is obtained by (1). Once the calculations for the 6 heads are completed, the results from each head are concatenated and mapped to the vector dimension set by the model using the W_O mapping.

D. MODEL ARCHITECTURE

Firstly, the original image is divided into 16 non-overlapping image patches. Then, ViT-Small is used to extract features from these 16 patches individually. After the extraction, the "classification token" from each of these 16 features are concatenated together to form a new feature $X_{Features} \in \mathbb{R}^{16\times384}$. Next, a learnable position embedding $E_{pos} \in \mathbb{R}^{16\times384}$ is added to this feature as the input to the GICB, as shown in (3). The classification results can be obtained by

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

passing this input through two GICBs and a simple fully connected layer. The two GICBs are determined by (4)-(7).

$$X = X_{Features} + E_{pos} \quad X \in \mathbb{R}^{16\times384} \quad (3)$$

$$A_1 = \mathrm{MSA}_1\big(\mathrm{LN}(X)\big) + X \quad A_1 \in \mathbb{R}^{16\times384} \quad (4)$$

$$M_1 = \mathrm{MLP}_1(A_1) \quad M_1 \in \mathbb{R}^{16\times128} \quad (5)$$

$$A_2 = \mathrm{MSA}_2\big(\mathrm{LN}(M_1)\big) + M_1 \quad A_2 \in \mathbb{R}^{16\times128} \quad (6)$$

$$M_2 = \mathrm{MLP}_2(A_2) \quad M_2 \in \mathbb{R}^{16\times1} \quad (7)$$

where the dimension of MSA_1 is 384, and it has 6 heads. MSA_2 has a dimension of 128 and 4 heads.

IV. EXPERIMENTS AND RESULTS

A. DATASETS AND EXPERIMENTAL SETUP

In this study, the APTOS [22] and Messidor-1 [23] datasets were used to test the proposed model. The APTOS consists of 5,590 color fundus retinal images. However, only 3,662 images have corresponding DR lesion grades, and were divided into five classes. Thus, only these labeled 3,662 color fundus retinal images in APTOS were used for training and testing the model in order to implement five-class classification of DR. The Messidor-1 contains 1,200 color fundus retinal images, which were specified as four classes [24, 25]. Due to the limited number of images in the Messidor-1 and the insufficient samples of certain class, the samples with annotations of grade 0 and 1 were taken as normal samples, while grade 2 and grade 3 were considered as referable DR. That is binary classification was performed on Messidor-1.

In the APTOS, 10% of the fundus retinal images were randomly selected as the test set. 80% of the remaining images were randomly chosen as the training set, and 20% were used as the validation set. All images were cropped to remove black regions and uniformly resized to 896×896 pixels. Image augmentation included random horizontal flips and random rotations within the range of -180 degrees to +180 degrees. The optimizer was AdamW with a weight decay of 0.3. The maximum learning rate was set to 1×10^{-4}, with a learning rate warm-up for the first 10 epochs. The learning rate was reduced by a factor of 0.5 at epochs 20, 35, and 60. The maximum number of epochs was set to be 110. The loss function was the cross-entropy loss with label smoothing of 0.05 [26]. For the Messidor-1, the data set partitioning and hyperparameter setting were similar to that with the APTOS. The binary cross-entropy loss function was used for DR prediction, and the maximum learning rate was set to be 1×10^{-5}.

All experiments in this study were conducted on an NVIDIA RTX 3090 GPU with 24GB of VRAM. The model with the highest accuracy on the validation set was saved during training and tested on the test set. The main evaluation metrics include accuracy, AUC, Kappa score and sensitivity. To avoid the randomness during model training, all experiments were repeated five times and the results were averaged. Additionally, all models were set to evaluation mode with autograd computation disabled and the batch size was uniformly set to be 16 in comparing the inference time of the models.

B. ABLATION EXPERIMENT

The experimental results on APTOS dataset are presented in Table 1. If the images are firstly compressed to 224×224 and then inputted into ViT, the ACC, AUC, Kappa and sensitivity of ViT (224) are 83.5%, 94.6%, 88.5% and 67.3%, respectively. When the high-resolution images were directly input into ViT for classification, the ACC, AUC, Kappa and sensitivity of ViT (896) are 83.9%, 94.6%, 90.3% and 68.3%, respectively. All the performance evaluation metrics of ViT (869) has been improved compared that of ViT (224). The high-resolution images with more information are conducive to mitigate overfitting, which takes a positive impact on the model's performance. However, 2940 position embedding weights were missing in directly inputting the 896×896 images into the pre-trained ViT because the sequence lengths of 896×896 and 224×224 images are respectively 3136 and 196. Although missing weights can be obtained through interpolation, when there are too many missing weights, the interpolated weights may have a negative impact on the model's classification performance. The present results indicate that the positive impact of high-resolution images is larger than the negative influence of weight interpolation.

It can be also found that the multiple instance learning approach without GICB and with a simple fully connected layer for classification (TMILv1) doesn't improve the model's prediction accuracy and only slightly improves the AUC. However, the multiple instance learning approach with GICB (TMILv2) significantly improves the classification performance. The accuracy, Kappa score and sensitivity of TMILv2 are 85.3%, 91.5% and 73.2%, respectively. In TMILv3, the positional relationship between instances was

TABLE 1. THE RESULT OF ABLATION STUDY ON APTOS DATASET (MEAN ± STANDARD DEVIATION, UNIT: %)

Model	Improvement			DR Classification			
	GICB	Position Embedding	Residual Connection	ACC	AUC	Kappa	Sen
ViT (224)	×	×	×	83.5±0.8	94.6±0.4	88.5±1.6	67.3±2.2
ViT (896)	×	×	×	83.9±1.6	94.6±1.6	90.3±1.3	68.3±2.5
TMILv1	×	×	×	83.2±0.9	95.7±0.5	90.2±1.2	67.5±1.5
TMILv2	√	×	×	85.3±0.5	94.9±0.8	**91.5±0.7**	73.2±1.6
TMILv3	√	√	×	84.5±1.3	**96.0±0.9**	**91.5±0.8**	72.1±2.1
TMILv4	√	√	√	**85.6±1.4**	95.6±0.7	90.7±0.6	**73.7±1.6**

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

TABLE 2. THE RESULT OF ABLATION STUDY ON MESSIDOR-1 DATASET (MEAN ± STANDARD DEVIATION, UNIT: %).

Model	Improvement			rDR Classification		
	GICB	Position Embedding	Residual Connection	ACC	AUC	Sen
ViT (224)	×	×	×	81±1.1	88.5±1.5	76.7±2.3
ViT (896)	×	×	×	87.5±2.6	93.8±1.2	78.1±2.6
TMILv1	×	×	×	91.2±2.0	95.9±0.6	83.7±5.0
TMILv2	√	×	×	92.2±2.0	**97.4±0.5**	84.1±4.4
TMIL v3	√	√	×	**93.1±2.2**	97±0.3	**86.9±4.0**
TMIL v4	√	√	√	**93.1±0.9**	97.2±0.4	**86.9±2.3**

taken into account and the learnable position embeddings were introduced after extracting features with ViT to facilitate the GICB's utilization of positional information. This improvement allows the model to maintain a Kappa score of 91.5% while further increases the AUC to 96.0%. In TMILv4, the residual connections were introduced in GICB, which further improves the classification performance. It can be seen that the accuracy, AUC, Kappa score and sensitivity of TMILv4 reach up to 85.6%, 95.6%, 90.7% and 73.7%, respectively. A brief analysis on the DR classification at various levels for both TMILv4 and ViT was also conducted. The confusion matrices for ViT (224), ViT (896) and TMILv4 on APTOS dataset are summarized in Table 3. Compared with ViT, TMILv4 indicates a slight decrease in sensitivity for DR level 2, while shows improvement in sensitivity for DR levels 1, 3 and 4.

TABLE 3. CONFUSION MATRIX ON APTOS.

Predicted label

True Label	0	1	2	3	4		
0	177	2	1	0	0	98.33%	1.67%
1	4	19	14	0	0	51.35%	48.65%
2	1	11	83	2	2	83.84%	16.16%
3	0	0	7	11	1	57.89%	42.11%
4	0	2	10	3	14	48.28%	51.72%
	97.25%	55.88%	72.17%	68.75%	82.35%		
	2.75%	44.12%	27.83%	31.25%	17.65%		

ViT (224)

Predicted label

True Label	0	1	2	3	4		
0	180	0	0	0	0	100%	0%
1	4	16	17	0	0	43.24%	56.76%
2	1	8	83	6	1	83.84%	16.16%
3	0	0	5	13	1	68.42%	31.58%
4	0	1	12	3	13	44.83%	55.17%
	97.3%	64%	70.94%	59.09%	86.67%		
	2.7%	36%	29.06%	40.91%	13.33%		

ViT (896)

Predicted label

True Label	0	1	2	3	4		
0	178	2	0	0	0	98.89%	1.11%
1	2	23	10	0	2	62.16%	37.84%
2	0	9	79	6	5	79.8%	20.2%
3	0	0	3	14	2	73.68%	26.32%
4	0	0	8	3	18	62.07%	37.93%
	98.89%	67.65%	79%	60.87%	66.67%		
	1.11%	32.35%	21%	39.13%	33.33%		

TMILv4

Subsequently, the model's binary classification performance was tested on the Messidor-1 dataset, and the results are summarized in Table 2. It is consistent with that on APTOS dataset that inputting higher-resolution retinal images into ViT improves the classification performance. Compared with the binary classification of ViT (224), the accuracy of

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

ViT (896) is increased from 81% to 87.5%, and AUC and sensitivity are enhanced to 93.8% and 78.1%, respectively. Using the multiple instance learning approach without the GICB and a single-layer fully connected layer for classification (TMILv1) based on Messidor-1 dataset further increases the accuracy, AUC and sensitivity to 91.2%, 95.9% and 83.7% respectively. While, the accuracy and AUC can be further increased to 92.2% and 97.4% by introducing GICB module in the multiple instance learning approach (TMILv2). Although the introduction of position embeddings (TMILv3) doesn't result in an improvement in AUC, the accuracy and sensitivity can be increased to 93.1% and 86.9%, respectively. Only AUC is slightly increased to 97.2% by introducing residual connections in the GICB (TMILv4). The confusion matrices for ViT (224), ViT (896) and TMILv4 on MESSIDOR dataset are presented in Table 4. It suggests that utilizing high-resolution images can enhance classification performance. And TMIL demonstrates further improvement in classification performance by using high-resolution images.

TABLE 4. CONFUSION MATRIX ON MESSIDOR.

Predicted label

True Label	0	1		
0	57	12	82.61%	17.39%
1	11	38	77.55%	22.45%
	83.82%	76%		
	16.18%	24%		

ViT (224)

Predicted label

True Label	0	1		
0	67	2	97.1%	2.9%
1	12	37	75.51%	24.49%
	84.81%	94.87%		
	15.19%	5.13%		

ViT (896)

Predicted label

True Label	0	1		
0	67	2	97.1%	2.9%
1	6	43	87.76%	12.24%
	91.78%	95.56%		
	8.22%	4.44%		

TMILv4

C. COMPARISON TO STATE-OF-THE-ART METHODS

The DR classification result of TMILv4 on APTOS dataset was compared with that by state-of-the-art methods. Vives-Boix *et al.* [27] assessed the capability of CNN models for DR classification on APTOS dataset. They showed that ResNet50 achieves an accuracy of 71.4% and sensitivity of 62.3%, while VGG-16 demonstrates an accuracy of 72.7% and sensitivity of 65.9%. And they improved ResNet50 and VGG16, and proposed AmResNet50 and AmVGG16 by enhancing the weight updating mechanism during the backpropagation process of CNN. The accuracy and sensitivity of AmResNet50 are 72.7% and 65.4%, while the accuracy and sensitivity of AmVGG16 are 78.6% and 72%, respectively. Although the sensitivity of AmVGG16 is close to that of TMIL, the accuracy of AmVGG16 is still much lower than that of TMIL. Mao *et al.* [28] modified the decoder structure of MAE and introduced a contrastive learning task to propose SDMAE. They achieved an accuracy of 83.1% on the APTOS dataset. Yue *et al.* [29] introduced ADCNet, which utilizes attention mechanisms to capture a significant amount of lesion perception information. They subsequently employ an attention-driven aggregation strategy to acquire a large number of DR-related features. Their ADCNet achieved an accuracy of 83.4%, an AUC of 94.26%, a Kappa score of 74.8%, and a sensitivity of 67.7% on the APTOS dataset, with overall classification performance slightly lower than TMILv4. The Texture Attention Network proposed by Alahmadi [30] achieves an accuracy of 85.1% and a Kappa score of 88.1% on the APTOS dataset. Feng *et al.* [31] proposed a graph neural network-based model and achieved an accuracy of 85.5% and a Kappa score of 90.6% on the APTOS dataset. While the proposed TMILv4 method achieves an accuracy of 85.6% and a Kappa score of 90.7%. And also, the accuracy of the proposed method surpasses that of the MIL-VT method proposed by Yu *et al.* [15]. It should be noted that MIL-VT, in addition to utilizing ImageNet pre-trained weights, also undergoes pre-training on over 340,000 retinal images. While, only the ImageNet pre-trained weights were required for TMILv4. The higher Kappa score of the MIL-VT method may be attributed to the fact that the weights pre-trained on a large-scale dataset of retinal images are better than that pre-trained on natural images.

TABLE 5. DR CLASSIFICATION RESULTS BY DIFFERENT METHODS ON APTOS DATASET (UNIT: %)

Model	ACC	AUC	Kappa	Sen
VGG-16 [27]	72.7	-	-	65.9
ResNet50 [27]	71.4	-	-	62.3
AmVGG16 [27]	78.6	-	-	72
AmResNet50 [27]	72.7	-	-	65.4
TLSPOFECOC [32]	82.1	-	-	82.1
SDMAE [28]	83.1	-	-	-
ADCNet [29]	83.4	94.26	74.8	67.7
Texture Attention Network [30]	85.1	-	88.1	**92**
Model based on GNN [31]	85.5	-	90.6	-
MIL-VT [15]	85.5	**97.9**	**92**	-
TMILv4	**85.6**	95.6	90.7	73.7

The proposed TMILv4 method on Messidor-1 dataset achieves highest accuracy. Mukherjee *et al.* [33] performed unaltered classical CNN models on Messidor-1 dataset. Similar to observations by Vives-Boix *et al.* [27], the DR classification performance of unmodified CNN models was generally low. The accuracy of Xception and MobileNet-V3_large are 80.3% and 81.9%, respectively. Hajabdollahi *et al.* [34] simplified the network structure and reduced network parameters through layer pruning based on VGG-16. The modified VGG-16 achieves an accuracy of 92% and an AUC

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

of 96.1%. Both of these metrics were lower than those achieved by TMILv4. And the AUC of TMILv4 are higher than that of Zoom-in-net method proposed by Wang *et al.* [35], CANet method proposed by Li *et al.* [36] and AFN method proposed by Lin *et al.* [37]. Sun *et al.* [14] did not report the accuracy of LAT on the Messidor-1 dataset. Although a higher AUC of LAT was reported, the DR classification needs to be used in conjunction of lesion detection. In comparison, the proposed TMIL focuses on DR classification and has a relatively simple structure.

TABLE 6. REFERABLE DR CLASSIFICATION RESULTS BY DIFFERENT METHODS ON MESSIDOR-1 DATASET (UNIT: %)

Model	ACC	AUC
ML-CAD [38]	75	72
Xception [33]	80.3	91.3
MobileNet-V3-Large [33]	81.9	94.1
Zoom-in-net [35]	91.1	95.7
Modified VGG-16 [34]	92	96.1
CANet [36]	92.6	96.3
AFN [37]	-	96.8
LAT [14]	-	**98.7**
TMILv4	**93.1**	97.2

D. COMPARISON OF INFERENCE TIME BETWEEN VIT AND TMIL ON HIGH-RESOLUTION IMAGES

TABLE 7. VIT AND TMIL INFERENCE TIME ON DIFFERENT DATASETS

Model	Dataset	Inference time (unit: second).
ViT	APTOS test set	12.33
TMIL	APTOS test set	4.71
ViT	Messidor-1 test set	3.90
TMIL	Messidor-1 test set	1.50

The inference time of TMIL and ViT on high-resolution images were tested under the same conditions. And the inference time on different datasets are shown in Table 7. It demonstrates that the proposed TMIL achieves higher classification performance on high-resolution retinal images while reduces inference time by 62%. This indicates that the proposed TMIL has significant advantages over the original ViT in fast patient screening.

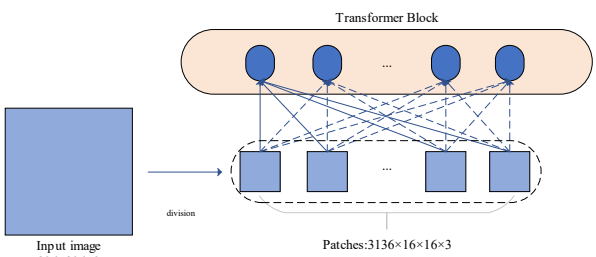


FIGURE 4. Schematic of ViT self-attention computation, where each vector is computed with every vector. This diagram omits steps such as vector mapping and MLP.

The faster inference time achieved by TMIL is attributed to the reduced sequence length resulting from the multiple instance learning approach.

More specifically, in (1), Q, K, and V are all matrices with dimensions $n \times d$. Here, n represents the sequence length, and d denotes the vector dimension (in the ViT-Small model used in this study, d corresponds to 384). Formula 1 requires computations between an $n \times d$ matrix and a $d \times n$ matrix to obtain an $n \times n$ matrix, which is subsequently operated on with an $n \times d$ matrix. This results in a time complexity of $O(n^2 d)$, where the computational complexity is squared relative to the sequence length.

In this study, with input images of size 896×896, as illustrated in Figure 4, directly inputting into ViT produces a sequence length of 3136. The Transformer must perform computations on 3136 vectors.

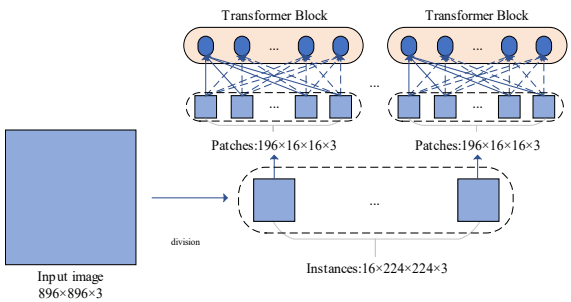


FIGURE 5. Schematic of TMIL self-attention computation, where the image is divided into 16 instances and the Transformer computes only within each instance. This diagram omits steps such as vector mapping and MLP.

In TMIL, as shown in Figure 5, an 896×896 image is divided into 16 instances, each of size 224×224. The Transformer only needs to process individual instances, with a significantly reduced sequence length of only 196 for each instance.

As a result, the reduced sequence length of 196, compared to the original 3136 sequence length, leads to a substantial reduction in inference time.

V. CONCLUSION

In this work, a Transformer-based multiple instance model is developed for diabetic retinopathy classification on high-resolution fundus retinal images. The proposed TMIL adopts the concept of multiple instance learning, where the high-resolution images are divided into several instances and ViT is employed to extract features within each instance. Then a Global Instance Computation Block (GICB) is designed to compute features from different instances and obtain global information by using attention mechanisms. Testing on the publicly available APTOS and Messidor-1 datasets demonstrates that the proposed TMIL model significantly reduces inference time and enhances disease classification accuracy compared to the original ViT. This suggests that the proposed multiple instance approach can be directly applied on high-resolution medical images and avoids the loss of lesion details caused by image overcompression. Furthermore,

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

it can alleviate the computational demands of ViT caused by excessively long sequences. However, it should be pointed out that there still exists a false-negative rate exceeding 30% of TMIL, although the sensitivity of TMIL for DR levels 1 and 4 is enhanced. The image augmentation can be employed to enhance lesion features and thereby enhance the sensitivity in detecting DR levels 1 and 4 in the future. And continuing efforts can be made on refining the GICB in order to take better advantage of global information and achieve higher disease classification performance. Additionally, the application of the proposed multiple instance method on 3D medical images will be another research topic.

REFERENCES

[1] N. H. Cho *et al.*, "IDF Diabetes Atlas: Global estimates of diabetes prevalence for 2017 and projections for 2045," *Diabetes Research and Clinical Practice*, vol. 138, pp. 271-281, 2018.

[2] R. Sarki, K. Ahmed, H. Wang, and Y. Zhang, "Automatic detection of diabetic eye disease through deep learning using fundus images: a survey," *IEEE Access*, vol. 8, pp. 151133-151149, 2020.

[3] N. Asiri, M. Hussain, F. Al Adel, and N. Alzaidi, "Deep learning based computer-aided diagnosis systems for diabetic retinopathy: A survey," *Artificial Intelligence in Medicine*, vol. 99, p. 101701, 2019.

[4] K. B. Nielsen, M. L. Lautrup, J. K. Andersen, T. R. Savarimuthu, and J. Grauslund, "Deep learning–based algorithms in screening of diabetic retinopathy: A systematic review of diagnostic performance," *Ophthalmology Retina*, vol. 3, no. 4, pp. 294-304, 2019.

[5] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[6] F. Shamshad *et al.*, "Transformers in medical imaging: A survey," *arXiv preprint arXiv:2201.09873*, 2022.

[7] N. S. Kumar and B. R. Karthikeyan, "Diabetic Retinopathy Detection using CNN, Transformer and MLP based Architectures," in *Proc. International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, 2021, pp. 1-2.

[8] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248-255.

[9] A. Krizhevsky and G. Hinton, "Learning multiple layers of features from tiny images," 2009.

[10] T.-Y. Lin *et al.*, "Microsoft coco: Common objects in context," in *Proc. Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Part V 13*, 2014, pp. 740-755.

[11] W. L. Alyoubi, M. F. Abulkhair, and W. M. Shalash, "Diabetic retinopathy fundus image classification and lesions localization system using deep learning," *Sensors*, vol. 21, no. 11, p. 3704, 2021.

[12] N. J. Mohan, R. Murugan, T. Goel, and P. Roy, "ViT-DR: Vision Transformers in Diabetic Retinopathy Grading Using Fundus Images," in *Proc. IEEE 10th Region 10 Humanitarian Technology Conference (R10-HTC)*, 2022, pp. 167-172.

[13] C. Matsoukas, J. F. Haslum, M. Söderberg, and K. Smith, "Is it time to replace cnns with transformers for medical images?" *arXiv preprint arXiv:2108.09038*, 2021.

[14] R. Sun, Y. Li, T. Zhang, Z. Mao, F. Wu, and Y. Zhang, "Lesion-aware transformers for diabetic retinopathy grading," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 10938-10947.

[15] S. Yu *et al.*, "Mil-vt: Multiple instance learning enhanced vision transformer for fundus image classification," in *Proc. Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Part VIII 24*, 2021, pp. 45-54.

[16] A. Myronenko, Z. Xu, D. Yang, H. R. Roth, and D. Xu, "Accounting for dependencies in deep learning based multiple instance learning for whole slide imaging," in *Proc. Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Part VIII 24*, 2021, pp. 329-338.

[17] Z. Qian *et al.*, "Transformer based multiple instance learning for weakly supervised histopathology image segmentation," in *Proc. Medical Image Computing and Computer Assisted Intervention–MICCAI 2022: 25th International Conference, Singapore, September 18–22, 2022, Part II*, 2022, pp. 160-170.

[18] T. G. Dietterich, R. H. Lathrop, and T. Lozano-Pérez, "Solving the multiple instance problem with axis-parallel rectangles," *Artificial Intelligence*, vol. 89, no. 1-2, pp. 31-71, 1997.

[19] V. Cheplygina, D. M. Tax, and M. Loog, "Multiple instance learning with bag dissimilarities," *Pattern Recognition*, vol. 48, no. 1, pp. 264-275, 2015.

[20] J. Amores, "Multiple instance classification: Review, taxonomy and comparative study," *Artificial Intelligence*, vol. 201, pp. 81-105, 2013.

[21] H. Li *et al.*, "DT-MIL: deformable transformer for multi-instance learning on histopathological image," in *Proc. Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Part VIII 24*, 2021, pp. 206-216.

[22] M. Karthik, Sohier Dane. APTOS 2019 Blindness Detection [Online] Available: https://kaggle.com/competitions/aptos2019-blindness-detection

[23] E. Decencière *et al.*, "Feedback on a publicly distributed image database: the Messidor database," *Image Analysis & Stereology*, vol. 33, no. 3, pp. 231-234, 2014.

[24] V. Gulshan *et al.*, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *Jama*, vol. 316, no. 22, pp. 2402-2410, 2016.

[25] C. P. Wilkinson *et al.*, "Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales," *Ophthalmology*, vol. 110, no. 9, pp. 1677-1682, 2003.

[26] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 2818-2826.

[27] V. Vives-Boix and D. Ruiz-Fernández, "Diabetic retinopathy detection through convolutional neural networks with synaptic metaplasticity," *Computer Methods and Programs in Biomedicine*, vol. 206, p. 106094, 2021.

[28] J. Mao, H. Zhou, X. Yin, Y. C. Xu, and B. N. Rui, "Masked autoencoders is an effective solution to transformer data-hungry," *arXiv preprint arXiv:2212.05677*, 2022.

[29] G. Yue, Y. Li, T. Zhou, X. Zhou, Y. Liu, and T. Wang, "Attention-Driven Cascaded Network for Diabetic Retinopathy Grading from Fundus Images," *Biomedical Signal Processing and Control*, vol. 80, p. 104370, 2023.

[30] M. D. Alahmadi, "Texture attention network for diabetic retinopathy classification," *IEEE Access*, vol. 10, pp. 55522-55532, 2022.

[31] M. Feng, J. Wang, K. Wen, and J. Sun, "Grading of Diabetic Retinopathy Images Based on Graph Neural Network," *IEEE Access*, 2023.

[32] W. Wong, F. H. Juwono, and C. Capriono, "Diabetic Retinopathy Detection and Grading: A Transfer Learning Approach Using Simultaneous Parameter Optimization and Feature-Weighted ECOC Ensemble," *IEEE Access*, 2023.

[33] N. Mukherjee and S. Sengupta, "Comparing deep feature extraction strategies for diabetic retinopathy stage classification from fundus images," *Arabian Journal for Science and Engineering*, pp. 1-20, 2023.

[34] M. Hajabdollahi, R. Esfandiarpoor, K. Najarian, N. Karimi, S. Samavi, and S. R. Soroushmehr, "Hierarchical pruning for simplification of convolutional neural networks in diabetic retinopathy classification," in *Proc. 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2019, pp. 970-973.

[35] Z. Wang, Y. Yin, J. Shi, W. Fang, H. Li, and X. Wang, "Zoom-in-net: Deep mining lesions for diabetic retinopathy detection," in *Proc. Medical Image Computing and Computer Assisted Intervention–*

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3351473

MICCAI 2017: 20th International Conference, Quebec City, QC, Canada, September 11-13, 2017, Part III 20, 2017, pp. 267-275.

[36] X. Li, X. Hu, L. Yu, L. Zhu, C.-W. Fu, and P.-A. Heng, "CANet: cross-disease attention network for joint diabetic retinopathy and diabetic macular edema grading," *IEEE Transactions on Medical Imaging*, vol. 39, no. 5, pp. 1483-1493, 2019.

[37] Z. Lin *et al.*, "A framework for identifying diabetic retinopathy based on anti-noise detection and attention-based fusion," in *Proc. Medical Image Computing and Computer Assisted Intervention–MICCAI 2018: 21st International Conference, Granada, Spain, September 16-20, 2018, Part II 11*, 2018, pp. 74-82.

[38] E. Abdelmaksoud, S. El-Sappagh, S. Barakat, T. Abuhmed, and M. Elmogy, "Automatic diabetic retinopathy grading system based on detecting multiple retinal lesions," *IEEE Access*, vol. 9, pp. 15939-15960, 2021.



Yaoming Yang was born in Yuhuan, Zhejiang, China in 1998. He received the B.E. degree in computer science and technology from the College of Modern Science and Technology, China Jiliang University, Hangzhou, China. He is currently pursuing the M.S. degree in electronic science and technology at China Jiliang University, College of Science, Hangzhou, China.

His research interests include image processing, machine learning, and artificial intelligence.



Zhili Cai was born in Shanghai, China in 1999. She received the B.E. degree in computer science and technology from Shanghai Business School, Shanghai, China in 2021. She is currently pursuing the M.S. degree in electronic information at China Jiliang University, College of Science, Hangzhou, China.

Her research interests include image processing, machine learning, artificial intelligence and porous media.



Shuxia Qiu was born in China in 1980. She received the B.S. degree in physics from Qufu Normal University, Qufu, China in 2004, the M.S. degree in theoretical physics from Huazhong University of Science and Technology, Wuhan, China in 2007 and the Ph.D. degree in engineering thermophysics from University of Shanghai for Science and Technology, Shanghai, China in 2021.

From 2009 to 2021, she was an engineer at China Jiliang University (CJLU), College of Science. Since 2022, she has been a senior engineer at CJLU. She is the author of more than 30 articles. Her research interest includes enhanced heat and mass transfer, porous media, CFD.



Peng Xu was born in China in 1981. He received the B.S. degree in physics from Qufu Normal University, and Ph.D. degree in condensed matter physics from Huazhong University of Science and Technology, Wuhan, China in 2008.

From 2007 to 2009, he was a visiting scholar with Transport Processes Research (TPR) group of National University of Singapore. And he was a visiting professor with Mine Multiphysics group of McGill University from Jun to Dec in 2015. Since 2018, he has been a Full Professor of Applied Physics at China Jiliang University (CJLU). He serves as the Deputy Dean of College of Science in CJLU and Editorial Board of FRACTALS. The research interests includes heat and mass transfer in porous media, image processing, multiscale and multiphysics modeling, fractal geometry and its applications in engineering, sludge dewatering and drying etc. He is the author of two books, one book chapter and more than 100 journal papers.

Dr. Xu was ranked as the World's Top 2% Scientists.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4